

10. Non-Deterministic Finite Automata

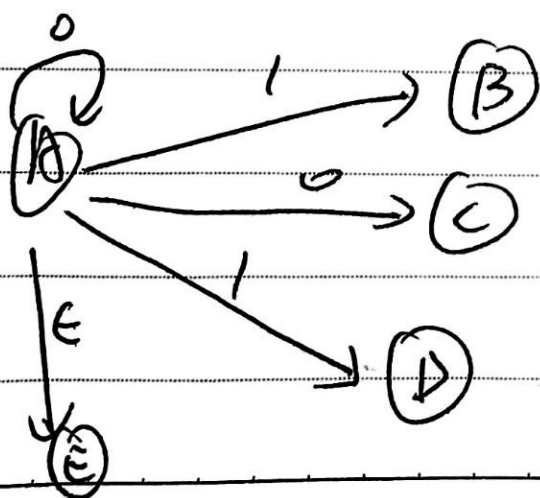
↓ ~~the~~ state

Deterministic Finite Automata

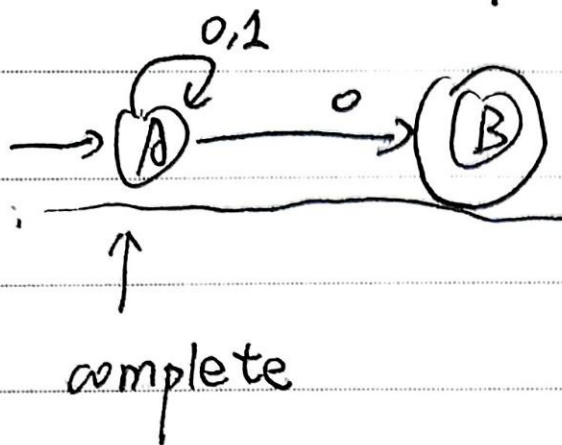
- ↳ In DFA, given the current state, we know what the next state will be
 - » It has only one unique next state
 - » It has no choices or randomness
 - » It is simple and easy to design

Non-Deterministic Finite Automata

- ↳ In NFA, given the current state, there could be multiple next states
 - » The next state may be chosen at random
 - » All the next states may be chosen in parallel



11. Formal Definition of NFA



$L = \{ \text{set of all strings that end with } 0 \}$

NFA definition $\{ Q, \Sigma, q_0, F, \delta \}$

- Q = set of all states - $\{A, B\}$
 - Σ = inputs - $\{0, 1\}$
 - q_0 = start state / initial state - A
 - F = Set of Final state - B
 - $\delta = Q \times \Sigma \rightarrow \underline{2^Q}$
- $\begin{cases} Q = \{A, B\} \\ \Sigma = \{0, 1\} \end{cases}$

$A \times 0 \rightarrow A$
 $A \times 0 \rightarrow B$
 $A \times 1 \rightarrow A$

$B \times 0 \rightarrow \phi$
 $B \times 1 \rightarrow \phi$

$A \xrightarrow{1} A, B, AB, \phi$

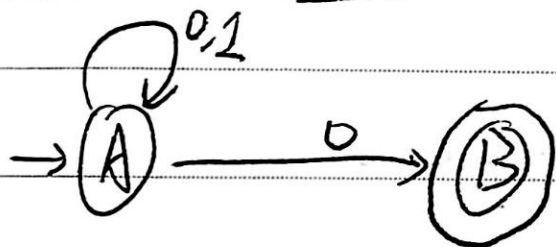
eg, three states. A, B, C

$A \xrightarrow{1} A, B, C, AB, AC, BC, \phi, ABC$

$\left\{ \begin{array}{l} 3 - 8 \text{ states - possibility} \\ 2 - 4 \end{array} \right.$

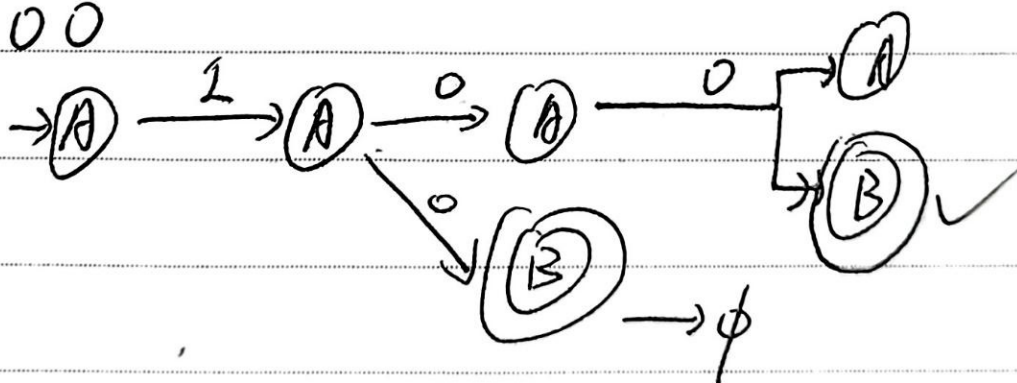
$\Rightarrow 2^2$ possibilities

12 NFA Examples

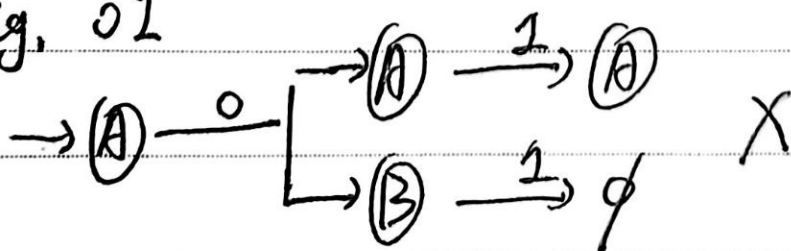


$L = \{ \text{set of all strings that end with 0} \}$

Eg. 100



$\Rightarrow$ Eg. 01

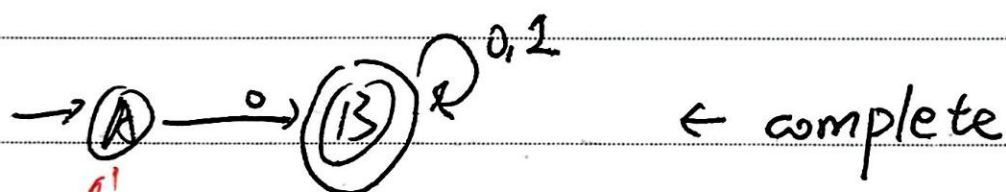


X

$\Rightarrow$ If there is any way to run the machine that ends in any set of states out of which at least one state is a final state, then the NFA accepts.

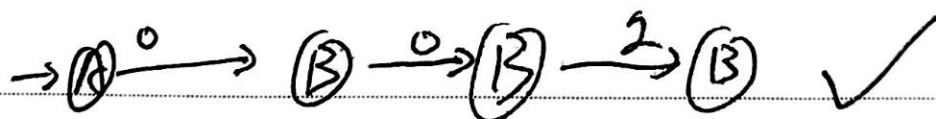
13. NFA - Example - 2

$L = \{ \text{Set of all strings that start with 0} \}$
 $= \{ 0, 00, 01, 000, \dots \}$

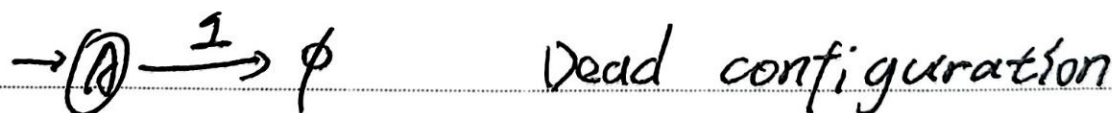


$\textcircled{C}$ 0, 1 (for DFA) (Dead State)

Eg. 001



Eg. 101



Eg. Construct a NFA that accepts sets of all strings over $\{0, 1\}$ of length 2

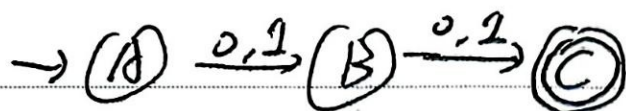


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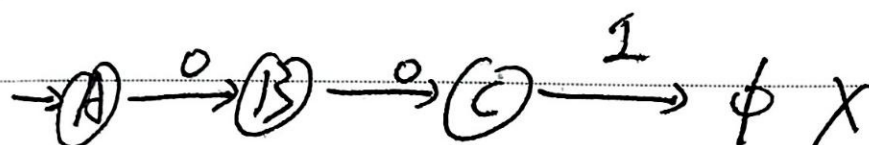
$\Sigma = \{0, 1\}$, $L = \{0001, 10, 11\}$



Eg. 00



Eg. 001

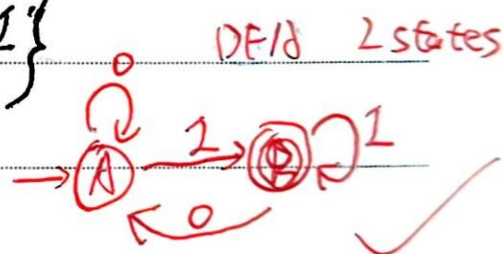
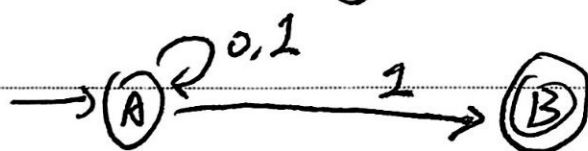


14, NFA - Example-3

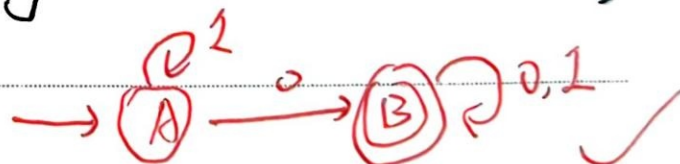
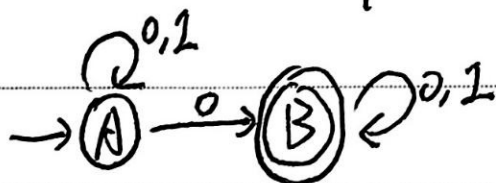
5.2

EX1).

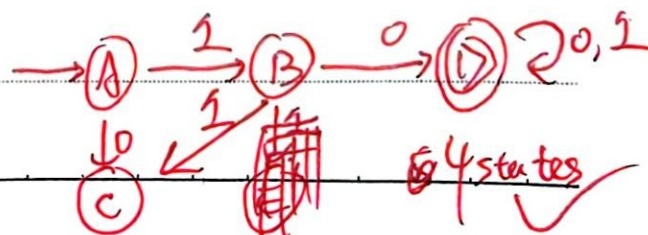
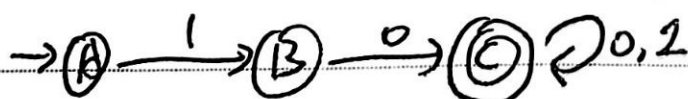
$L1 = \{\text{all strings that ends with '1'}\}$

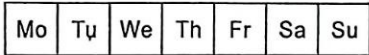


EX2) $L2 = \{\text{Set of all strings that contain '0'}\}$



EX3) $L3 = \{\text{Set of all strings that starts with '10'}\}$





Date / /

→ (A) $\xrightarrow{0}$ (B) $\xrightarrow{1}$ (C) $\xrightarrow{0.1}$

~~011~~ 001

A →⁰ A →⁰ A →¹ A
A →⁰ B →¹ C ✓
B →¹ ∅

→ A $\xrightarrow{0,1}$ A $\xrightarrow{2}$ B $\xrightarrow{1}$ C